

DATA ANALYTICS LEVEL III

Day 8: Introduction to Data Summarization & Pivot Tables

Student Activities

Day 8 — Student Activities Overview

Complete the following activities during your asynchronous session (3.5 hours total).

Unit of Competency: Summarize Data Sets (Core 3)

Learning Outcome Assessed: LO1 — Aggregate pivot table data

Activity	Type	Duration	Weight
Pivot Table Exercises (5 pivots)	Formative	1.5 hours	Graded (50 pts)
SQL Aggregation Parallel (5 queries)	Formative	0.75 hours	Graded (30 pts)
Reading: Pivot Tables In-Depth	Self-study	0.75 hours	N/A
Learning Journal Entry #8	Reflective	0.5 hours	Part of participation grade

Activity 1: Pivot Table Exercises

Type: Graded (50 points)

Time: 1.5 hours

File Needed: Day08_HR_Dataset.xlsx (download from LMS)

You are an HR analyst. Using the 50-employee dataset, build FIVE pivot tables that answer the questions below. Each pivot should live on its own sheet, named clearly (e.g., 'Pivot1_Headcount').

Pivot Table 1 — Headcount per Department (10 pts)

- **Rows:** department
- **Values:** Count of employee_id

- Question to answer in text below the pivot: Which department has the most employees? The fewest?

Pivot Table 2 — Total Salary Spend per City (10 pts)

- **Rows:** city
- **Values:** Sum of salary (format as Philippine Peso)
- **Question:** In which city does the company spend the most on salaries?

Pivot Table 3 — Average Salary per Position (10 pts)

- **Rows:** position
- **Values:** Average of salary (round to 0 decimals)
- Sort the pivot from highest to lowest average salary
- **Question:** Which position has the highest average salary? Is the difference between top and bottom large or small?

Pivot Table 4 — Department × City Cross-Tab (10 pts)

- **Rows:** department
- **Columns:** city
- **Values:** Count of employee_id
- **Question:** Which department–city combination has the most employees?

Pivot Table 5 — Hires per Year (10 pts)

- **Rows:** hire_date — then right-click → Group → check 'Years'
- **Values:** Count of employee_id
- **Question:** In which year did the company hire the most employees? Which years were slow hiring years?

Grading Rubric — Activity 1

Criterion	Points
Pivot 1 correct (headcount per dept)	10
Pivot 2 correct (total salary per city)	10
Pivot 3 correct (avg salary per position)	10
Pivot 4 correct (dept × city cross-tab)	10
Pivot 5 correct (hires per year, grouped)	10
Total	50 points

Save as: Day08_Pivots_[YourName].xlsx

Activity 2: SQL Aggregation Parallel

Type: Graded (30 points)

Time: 45 minutes

File Needed: Day08_practice.db (download from LMS)

Connect to Day08_practice.db in VS Code (SQLTools). Write SQL GROUP BY queries that produce the SAME summary as each of your 5 pivot tables above. This proves you understand both tools and can choose between them.

Queries to Write (6 points each)

1. Headcount per department, sorted from largest to smallest.
2. Total salary per city, sorted descending.
3. Average salary per position, rounded to 0 decimals, sorted descending.
4. Headcount per (department, city) combination.
5. Headcount per hire year. Hint: use `substr(hire\_date, 1, 4) AS hire\_year`.

Submission Format

For each query, write the question number, your SQL, and the first 3 rows of result as a comment:

```
-- Q1: Headcount per department
SELECT department, COUNT(*) AS headcount
FROM employees
GROUP BY department
ORDER BY headcount DESC;
-- Result (top 3):
-- IT          | 11
-- Sales       | 10
-- Finance     | 9
```

Grading Rubric — Activity 2

Criterion	Points
Q1 query correct + result matches pivot 1	6
Q2 query correct + result matches pivot 2	6
Q3 query correct + result matches pivot 3	6
Q4 query correct + result matches pivot 4	6
Q5 query correct + result matches pivot 5	6
Total	30 points

Save as: Day08_SQL_[YourName].sql (or .txt)

Learning Journal Entry #8

Time: 30 minutes

Reflect on today's session. Write at least 5 sentences addressing these prompts:

6. What is the difference between a pivot table and a regular Excel table?
7. When would you choose a pivot table over a SQL GROUP BY? When would you choose SQL?
8. Which of the 5 pivots was the trickiest to build? Why?
9. Did your pivot totals match your SQL totals exactly? If not, what was different?
10. Name one business question (real or imagined) you could answer with a pivot table tomorrow.

Save as: Day08_Journal_[YourName].docx

Upload all submissions to LMS before the next session.

Summary of Submissions

Submission	Filename	Due
Pivot Tables	Day08_Pivots_[YourName].xlsx	Before next session
SQL Aggregation	Day08_SQL_[YourName].sql	Before next session
Learning Journal	Day08_Journal_[YourName].docx	Before next session